Question 1

Answer saved

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Assume that the augmented matrix of the system of linear equations is given as follows:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 3 & 2 & -1 & 3 \\ -1 & -1 & 2 & -1 \end{array}\right]$$

Use the following steps to **reduce** the augmented matrix to **generalized row echelon form:**

Step 1:

$R_3 + R_1 \longrightarrow R_3$

$R_2 - 3R_1 \longrightarrow R_2$

Step 2:

$-R_2 \longrightarrow R_2$

- ○ $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$
- ○ $\left[\begin{array}{ccc|c} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array}\right]$
- ○ $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right]$
- ○ None of the given options
- ◉ $\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 1 & 1 \end{array}\right]$